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Course: Machine Learning Lab 1

Basic - Iris Flower dataset

1. Write a Python program to load the iris data from a given csv file into a dataframe and print the shape of the data, type of the data and first 3 rows.

```
from sklearn.datasets import load_iris
import pandas as pd
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
df['species'] = iris.target
df.to_csv('iris_data.csv', index=False)

print("Shape of the data:")
print(df.shape)
print("\nData Type:")
print(type(df))
print("\nFirst 3 rows:")
print(df.head(3))
```

```
Shape of the data:
(150, 5)

Data Type:
<class 'pandas.core.frame.DataFrame'>

First 3 rows:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2

   species
0        0
1        0
2        0
```

2. Write a Python program using Scikit-learn to print the keys, number of rows-columns, feature names and the description of the Iris data.

```
from sklearn.datasets import load_iris
import pandas as pd
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
df['species'] = iris.target
df.to_csv('iris_data.csv', index=False)

print("\nKeys of Iris dataset:")
print(df.keys())
print("\nNumber of rows and columns of Iris dataset:")
print(df.shape)
```

```
Keys of Iris dataset:
Index(['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)',
       'petal width (cm)', 'species'],
      dtype='object')

Number of rows and columns of Iris dataset:
(150, 5)
```

3. Write a Python program to get the number of observations, missing values and nan values.

```
from sklearn.datasets import load_iris
import pandas as pd
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
df['species'] = iris.target
df.to_csv('iris_data.csv', index=False)

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
 4   species            150 non-null    int64
dtypes: float64(4), int64(1)
memory usage: 6.0 KB
```

4. Write a Python program to create a 2-D array with ones on the diagonal and zeros elsewhere. Now convert the NumPy array to a SciPy sparse matrix in CSR format.

```
from scipy import sparse
import numpy as np
eye = np.eye(4)
print("NumPy array:\n", eye)
sparse_matrix = sparse.csr_matrix(eye)
print("\nSciPy sparse CSR matrix:\n", sparse_matrix)
```

```
NumPy array:
 [[1. 0. 0. 0.]
 [0. 1. 0. 0.]
 [0. 0. 1. 0.]
 [0. 0. 0. 1.]]

SciPy sparse CSR matrix:
   (0, 0)	1.0
  (1, 1)	1.0
  (2, 2)	1.0
  (3, 3)	1.0
```

5. Write a Python program to view basic statistical details like percentile, mean, std etc. of iris data.

```
from sklearn.datasets import load_iris
import pandas as pd
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
```

```
df['species'] = iris.target
df.to_csv('iris_data.csv', index=False)

df.describe()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333	1.000000
std	0.828066	0.435866	1.765298	0.762238	0.819232
min	4.300000	2.000000	1.000000	0.100000	0.000000
25%	5.100000	2.800000	1.600000	0.300000	0.000000
50%	5.800000	3.000000	4.350000	1.300000	1.000000
75%	6.400000	3.300000	5.100000	1.800000	2.000000
max	7.900000	4.400000	6.900000	2.500000	2.000000

6. Write a program to get observations of each species (setosa, versicolor, virginica) from iris data.

```
from sklearn.datasets import load_iris
import pandas as pd
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
df['species'] = iris.target
df.to_csv('iris_data.csv', index=False)

print("Observations of each species:")
print(df['species'].value_counts())
```

```
Observations of each species:
species
0    50
1    50
2    50
Name: count, dtype: int64
```

7. Write a Python program to drop species column from a given Dataframe and print the modified part. Call iris.csv to create the Dataframe.

```
from sklearn.datasets import load_iris
```

```
import pandas as pd
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
df['species'] = iris.target
df.to_csv('iris_data.csv', index=False)

print("Original Data:")
print(df.head())
new_data = df.drop('species',axis=1)
print("After removing id column:")
print(new_data.head())
```

```
Original Data:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   species
0        0
1        0
2        0
3        0
4        0
After removing id column:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2
```

8. Write a Python program to access first four cells from a given Dataframe using the index and column labels. Call iris.csv to create the Dataframe

```
from sklearn.datasets import load_iris
import pandas as pd
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
df['species'] = iris.target
df.to_csv('iris_data.csv', index=False)
```

```
print(df.head())
x = df.iloc[:, [1, 2, 3, 4]].values
print(x)
```

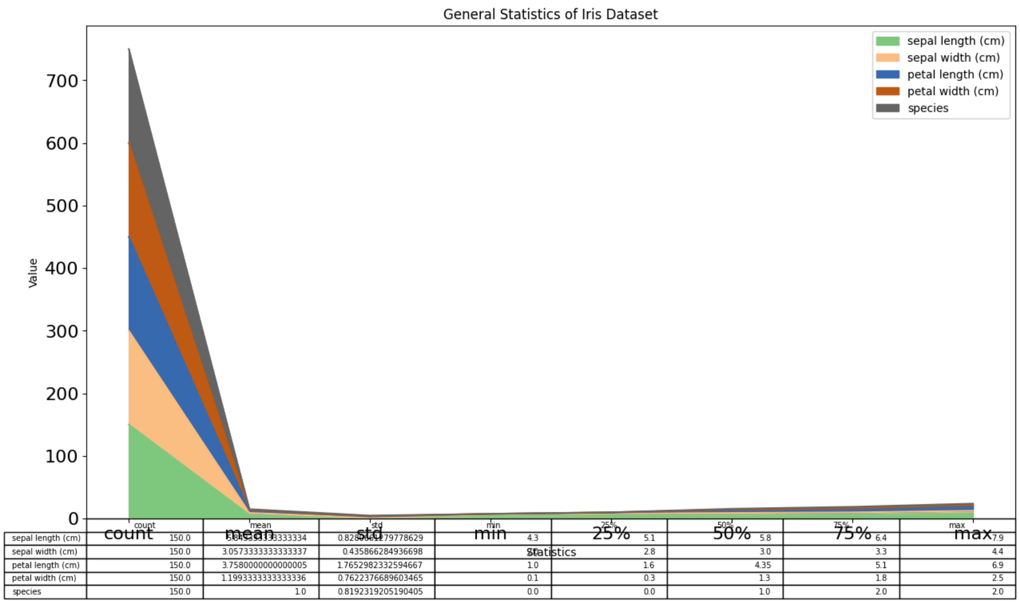
```
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   species
0        0
1        0
2        0
3        0
4        0
[[3.5 1.4 0.2 0. ]
 [3.  1.4 0.2 0. ]
 [3.2 1.3 0.2 0. ]
 [3.1 1.5 0.2 0. ]
 [3.6 1.4 0.2 0. ]
 [3.9 1.7 0.4 0. ]
 [3.4 1.4 0.3 0. ]
 [3.4 1.5 0.2 0. ]
 [2.9 1.4 0.2 0. ]
 [3.1 1.5 0.1 0. ]
 [3.7 1.5 0.2 0. ]
 [3.4 1.6 0.2 0. ]
 [3.  1.4 0.1 0. ]
 [3.  1.1 0.1 0. ]
 [4.  1.2 0.2 0. ]
 [4.4 1.5 0.4 0. ]
 [3.9 1.3 0.4 0. ]
 [3.5 1.4 0.3 0. ]
 [3.8 1.7 0.3 0. ]
 [3.8 1.5 0.3 0. ]
 [3.4 1.7 0.2 0. ]
 [3.7 1.5 0.4 0. ]
 [3.6 1.  0.2 0. ]
```

Visualization - Iris flower dataset

1. Write a Python program to create a plot to get a general Statistics of Iris data.

```
from sklearn.datasets import load_iris
import matplotlib.pyplot as plt
import pandas as pd
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
df['species'] = iris.target
df.to_csv('iris_data.csv', index=False)
```

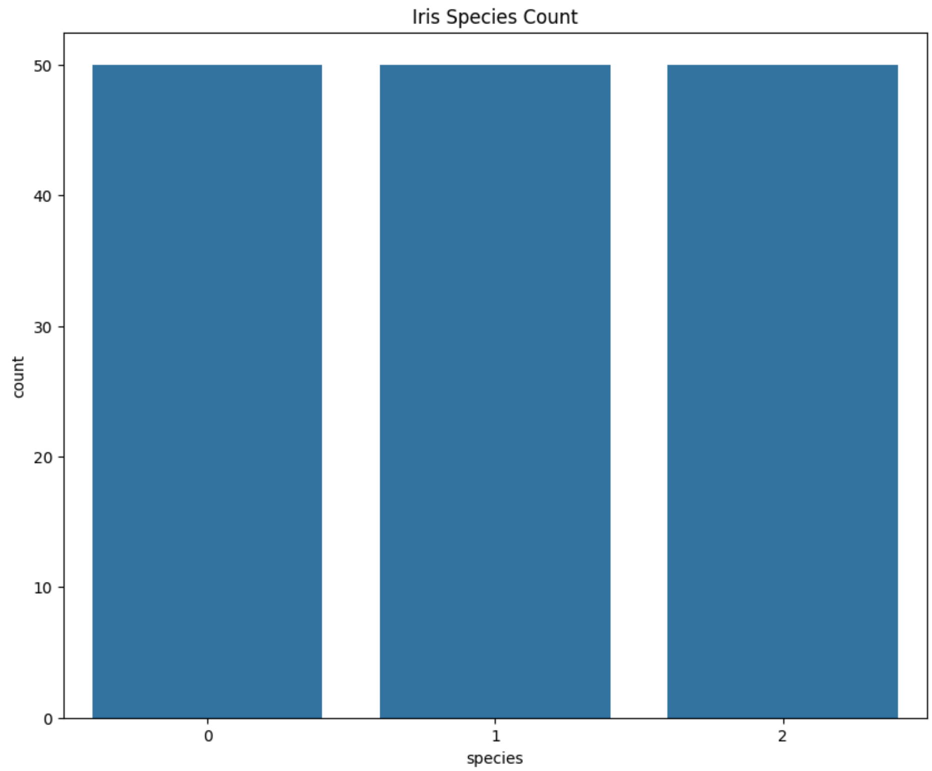
```
df.describe().plot(kind = "area",fontsize=16, figsize =
(15,8), table = True, colormap="Accent")
plt.xlabel('Statistics')
plt.ylabel('Value')
plt.title("General Statistics of Iris Dataset")
plt.show()
```



	count	mean	std	min	25%	50%	75%	max
sepal length (cm)	150.0	5.843333333333334	0.8280661279778629	4.3	5.1	5.8	6.4	7.9
sepal width (cm)	150.0	3.0573333333333337	0.435866284936698	2.0	2.8	3.0	3.3	4.4
petal length (cm)	150.0	3.7580000000000005	1.7652982332594667	1.0	1.6	4.35	5.1	6.9
petal width (cm)	150.0	1.1993333333333336	0.7622376689603465	0.1	0.3	1.3	1.8	2.5
species	150.0	1.0	0.8192319205190405	0.0	0.0	1.0	2.0	2.0

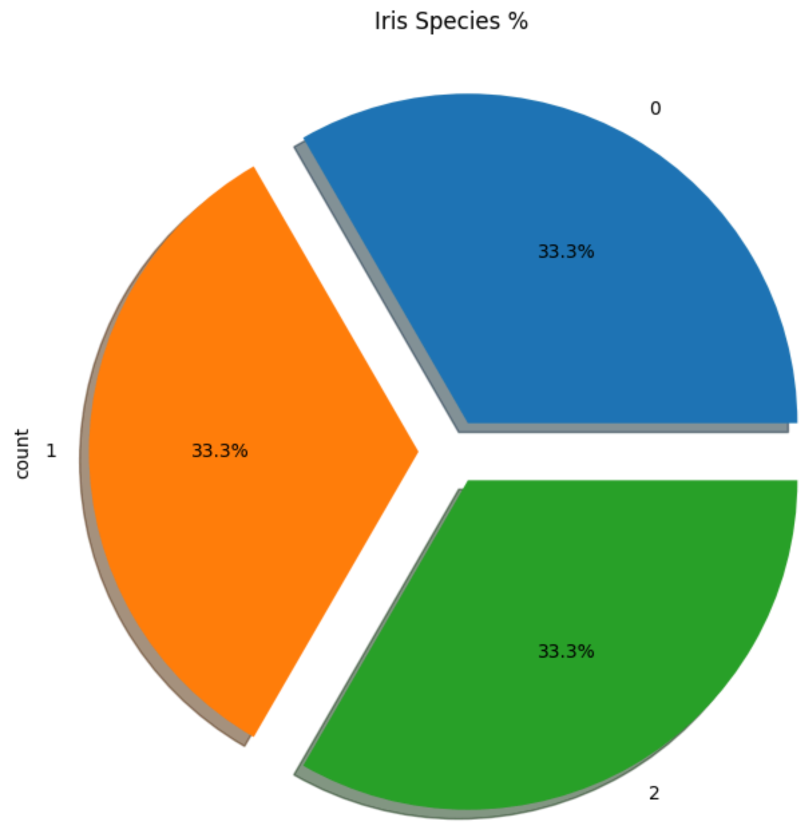
2. program to create a Bar plot to get the frequency of the three species of the Iris data.

```
from sklearn.datasets import load_iris
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
df['species'] = iris.target
df.to_csv('iris_data.csv', index=False)

ax=plt.subplots(1,1,figsize=(10,8))
sns.countplot(x='species',data=df)
plt.title("Iris Species Count")
plt.show()
```



3. Write a Python program to create a Pie plot to get the frequency of the three species of the Iris data.

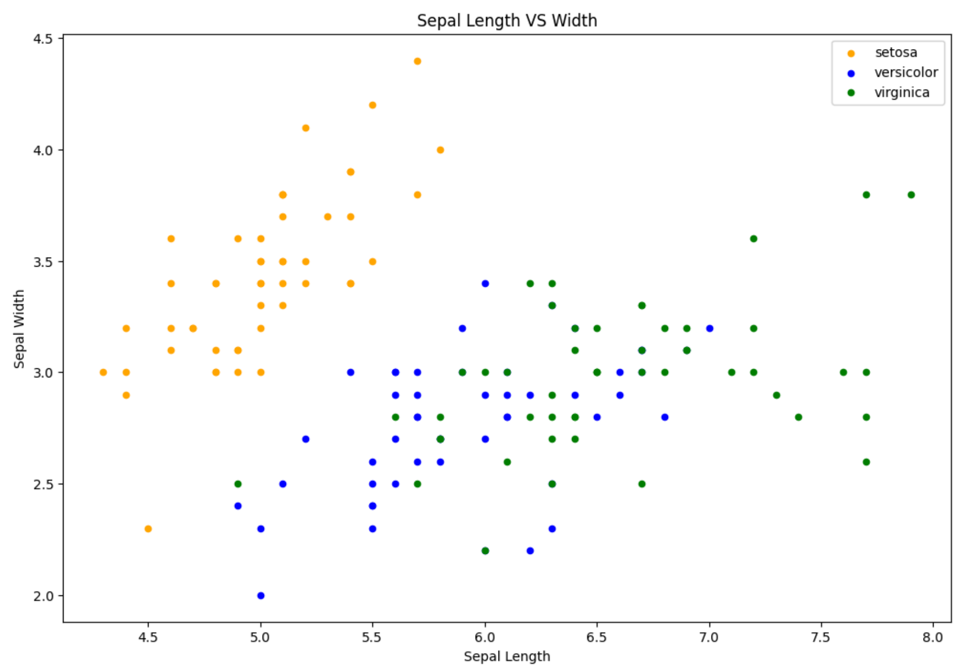
```
from sklearn.datasets import load_iris
import matplotlib.pyplot as plt
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
df['species'] = iris.target
df.to_csv('iris_data.csv', index=False)

ax=plt.subplots(1,1,figsize=(10,8))
df['species'].value_counts().plot.pie(explode=[0.1,0.1,0.1]
,autopct='%1.1f%%',shadow=True,figsize=(10,8))
plt.title("Iris Species %")
plt.show()
```



4. Write a Python program to create a graph to find relationship between the sepal length and width.

```
from sklearn.datasets import load_iris
import matplotlib.pyplot as plt
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
df['species'] = iris.target
df.to_csv('iris_data.csv', index=False)

fig = df[df.species==0].plot(kind='scatter',x='sepal length
(cm)',y='sepal width (cm)',color='orange', label='setosa')
df[df.species==1].plot(kind='scatter',x='sepal length
(cm)',y='sepal width (cm)',color='blue',
label='versicolor',ax=fig)
```

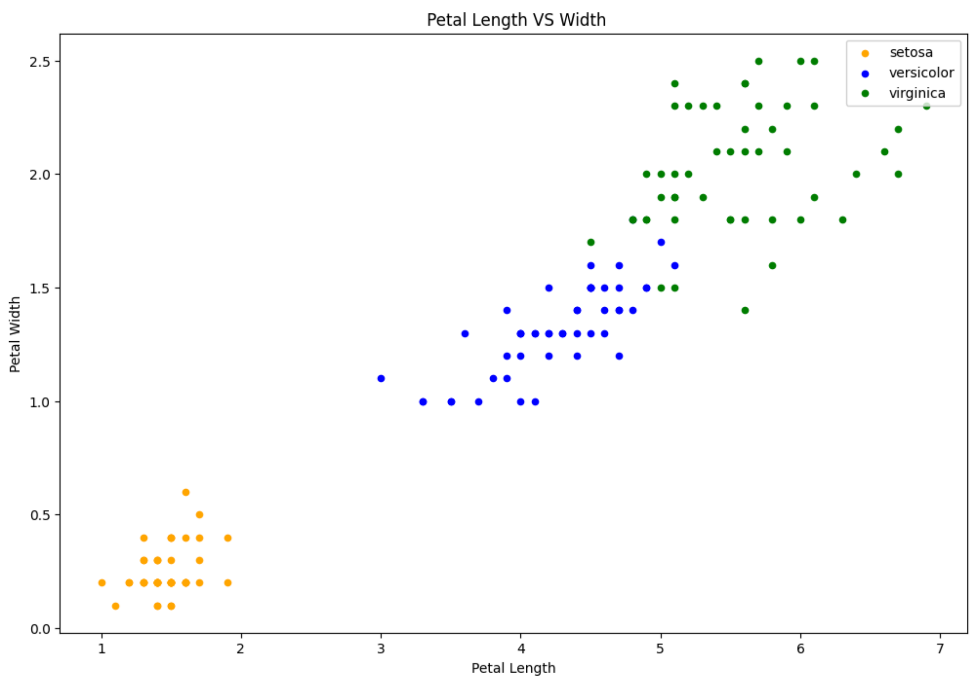
```
df[df.species==2].plot(kind='scatter',x='sepal length
(cm)',y='sepal width (cm)',color='green',
label='virginica', ax=fig)
fig.set_xlabel("Sepal Length")
fig.set_ylabel("Sepal Width")
fig.set_title("Sepal Length VS Width")
fig=plt.gcf()
fig.set_size_inches(12,8)
plt.show()
```



5. Write a Python program to create a graph to find relationship between the petal length and width.

```
from sklearn.datasets import load_iris
import matplotlib.pyplot as plt
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
df['species'] = iris.target
df.to_csv('iris_data.csv', index=False)

fig = df[df.species==0].plot(kind='scatter',x='petal length
(cm)',y='petal width (cm)',color='orange', label='setosa')
```

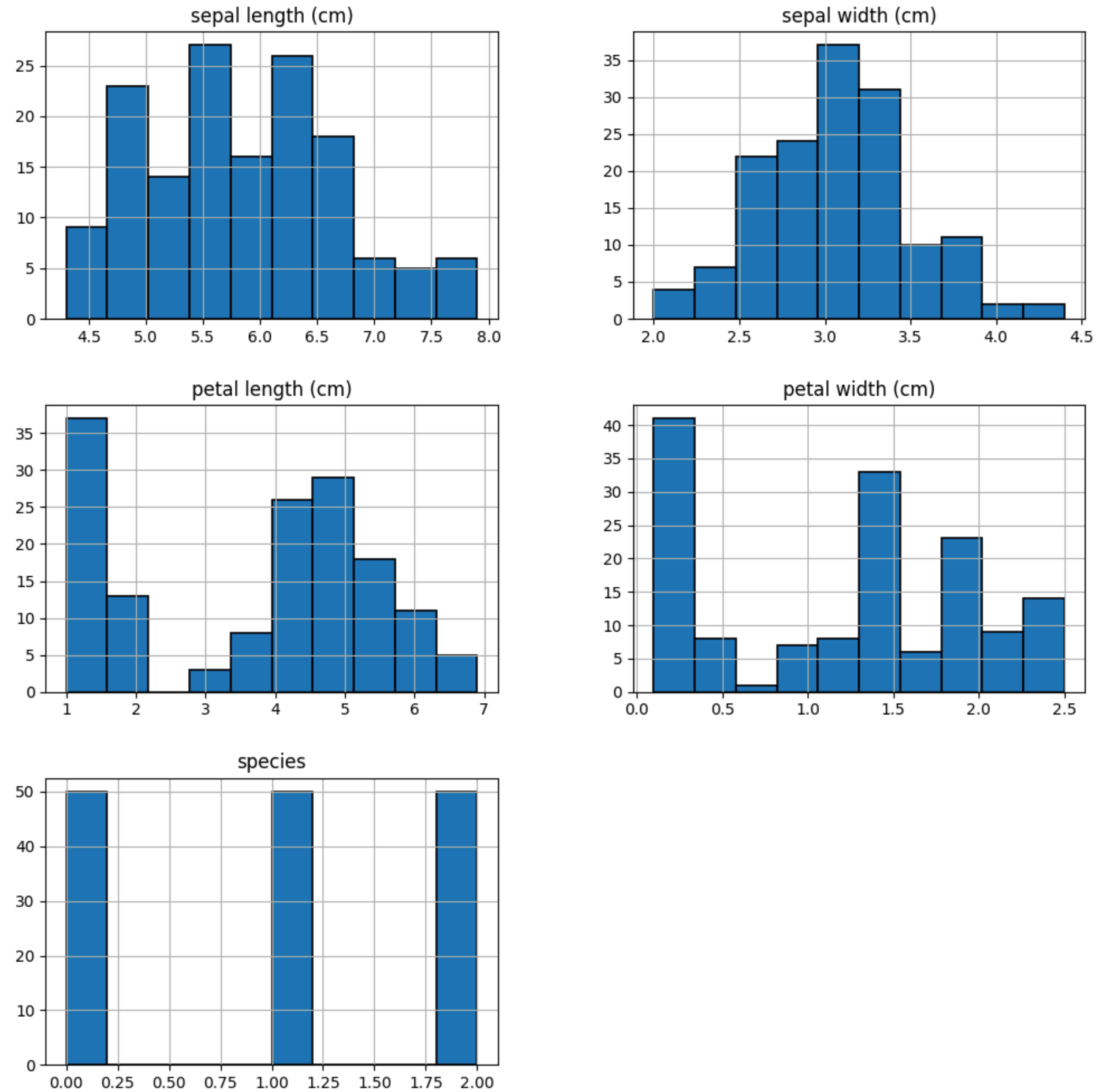
```
df[df.species==1].plot(kind='scatter',x='petal length
(cm)',y='petal width (cm)',color='blue',
label='versicolor',ax=fig)
df[df.species==2].plot(kind='scatter',x='petal length
(cm)',y='petal width (cm)',color='green',
label='virginica', ax=fig)
fig.set_xlabel("Petal Length")
fig.set_ylabel("Petal Width")
fig.set_title(" Petal Length VS Width")
fig=plt.gcf()
fig.set_size_inches(12,8)
plt.show()
```



6. Write a Python program to create a graph to see how the length and width of SepalLength, SepalWidth, PetalLength, PetalWidth are distributed.

```
from sklearn.datasets import load_iris
import matplotlib.pyplot as plt
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
df['species'] = iris.target
```

```
df.to_csv('iris_data.csv', index=False)

df.hist(edgecolor='black', linewidth=1.2)
fig=plt.gcf()
fig.set_size_inches(12,12)
plt.show()
```



7. Write a Python program to create a joinplot to describe individual distributions on the same plot between Sepal length and Sepal width.

```
from sklearn.datasets import load_iris
```

```
import matplotlib.pyplot as plt
import seaborn as sns
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
df['species'] = iris.target
df.to_csv('iris_data.csv', index=False)

fig=sns.jointplot(x='sepal length (cm)', y='sepal width
(cm)', data=df, color='blue')
plt.show()
```